Fashion Classifier AML

Automatic Clothing Category Recognition Using Classical Machine Learning

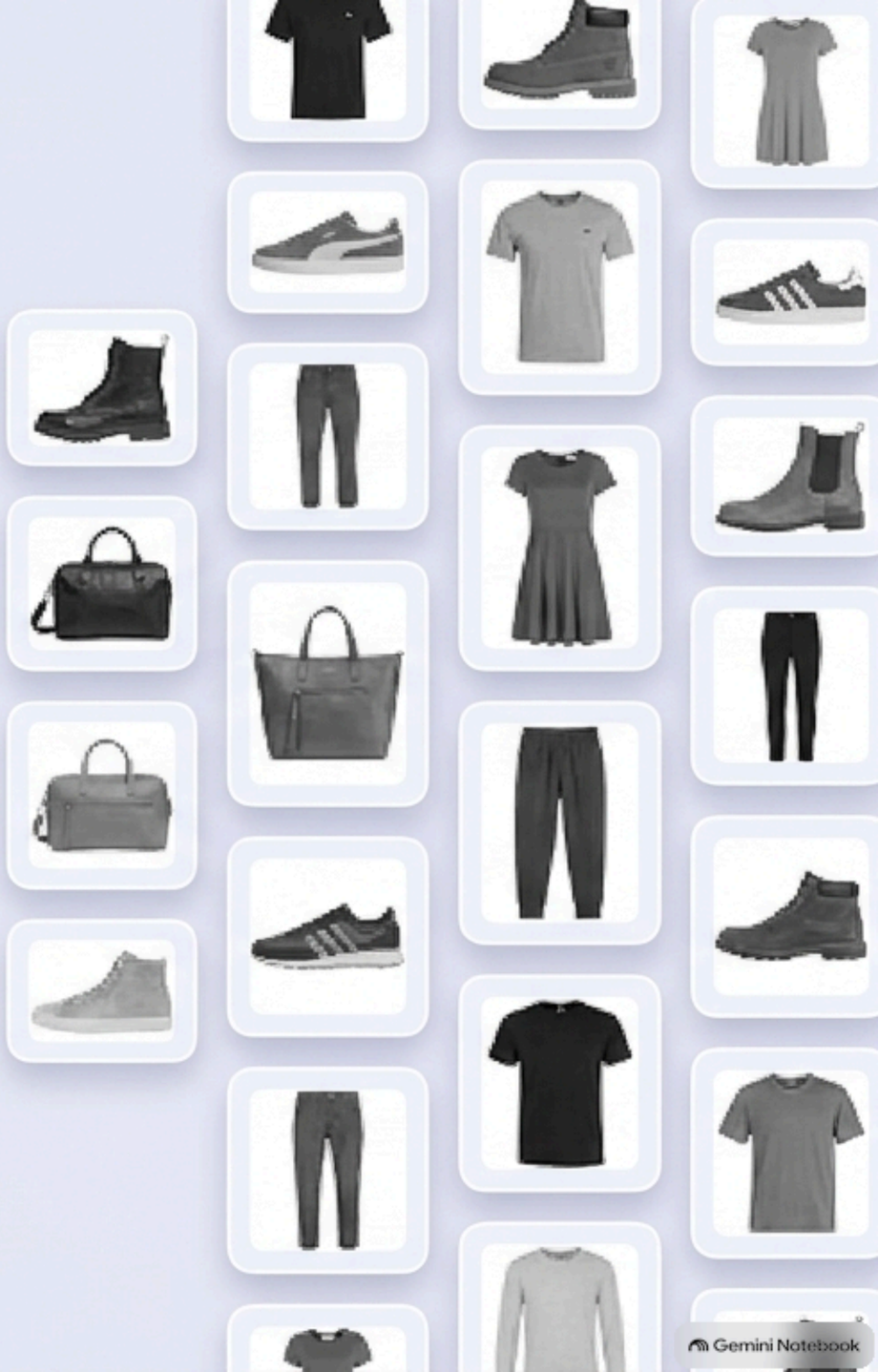
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Advanced Machine Learning Project (2024) | Newton School of Technology

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Project repository:
https://github.com/krishnaverma09/Fashion_Classifier_AML



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Benchmarking Classification on the Fashion-MNIST Dataset

The Challenge: Classifying 28×28 grayscale images lacking fine color details, where upper-body garments share overlapping silhouettes.

The Official Dataset: 60,000 training images, 10,000 test images.

Our Benchmark Budget: To prevent data leakage and evaluate algorithms fairly, we partitioned the official training data into a 48,000-image training pool and a 12,000-image validation set.

Constraint: All classical models were trained strictly on a shared subset of 12,000 images (1,200 per class).



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Feature Engineering: Raw Pixel Intensity vs. Edge Direction

Dataset → Preprocessing → Feature Extraction → ML Models → Prediction

Raw Pixels



784 Numerical Values

Preserves spatial locations. Normalized pixel intensities ranging from 0 to 1.

HOG (Histogram of Oriented Gradients)



324 Features

Summarizes local edge directions and outlines (9 bins, 7x7 cells, 2x2 blocks). Helps the classifier focus on garment shape rather than individual lighting.

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Evaluating Four Distinct Classical Machine Learning Approaches

Logistic Regression

Learns weighted combinations of features to estimate class probabilities.

[Fast Inference] Very low latency at prediction time.

k-Nearest Neighbors (k-NN)

Predicts a category by looking at the labels of the most similar training examples (k=5).

[Local Density Dependent] Requires comparing against the dataset during prediction.

Random Forest

Combines multiple decision trees to capture non-linear, complex patterns in the features.

[High Complexity Handling] Naturally adapts to intricate feature interactions.

Linear SVM

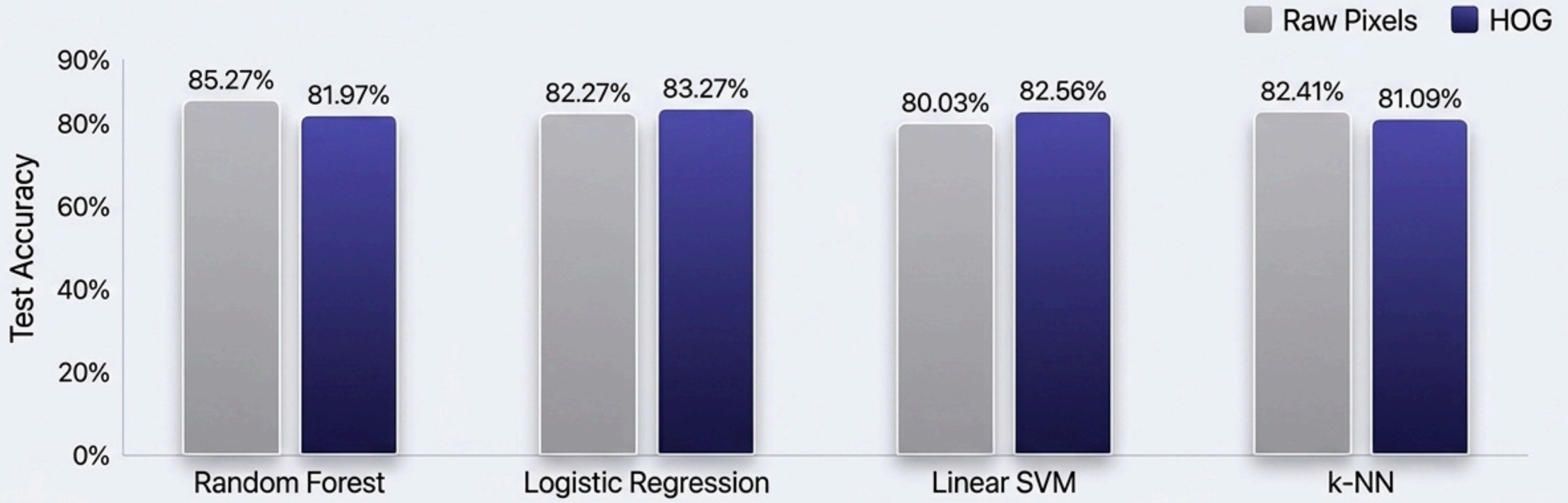
Learns decision boundaries that separate the 10 clothing categories with the largest possible margin.

[Clear Margins] Highly effective when data is linearly separable.

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Random Forest Dominates, While HOG Rescues Linear Models

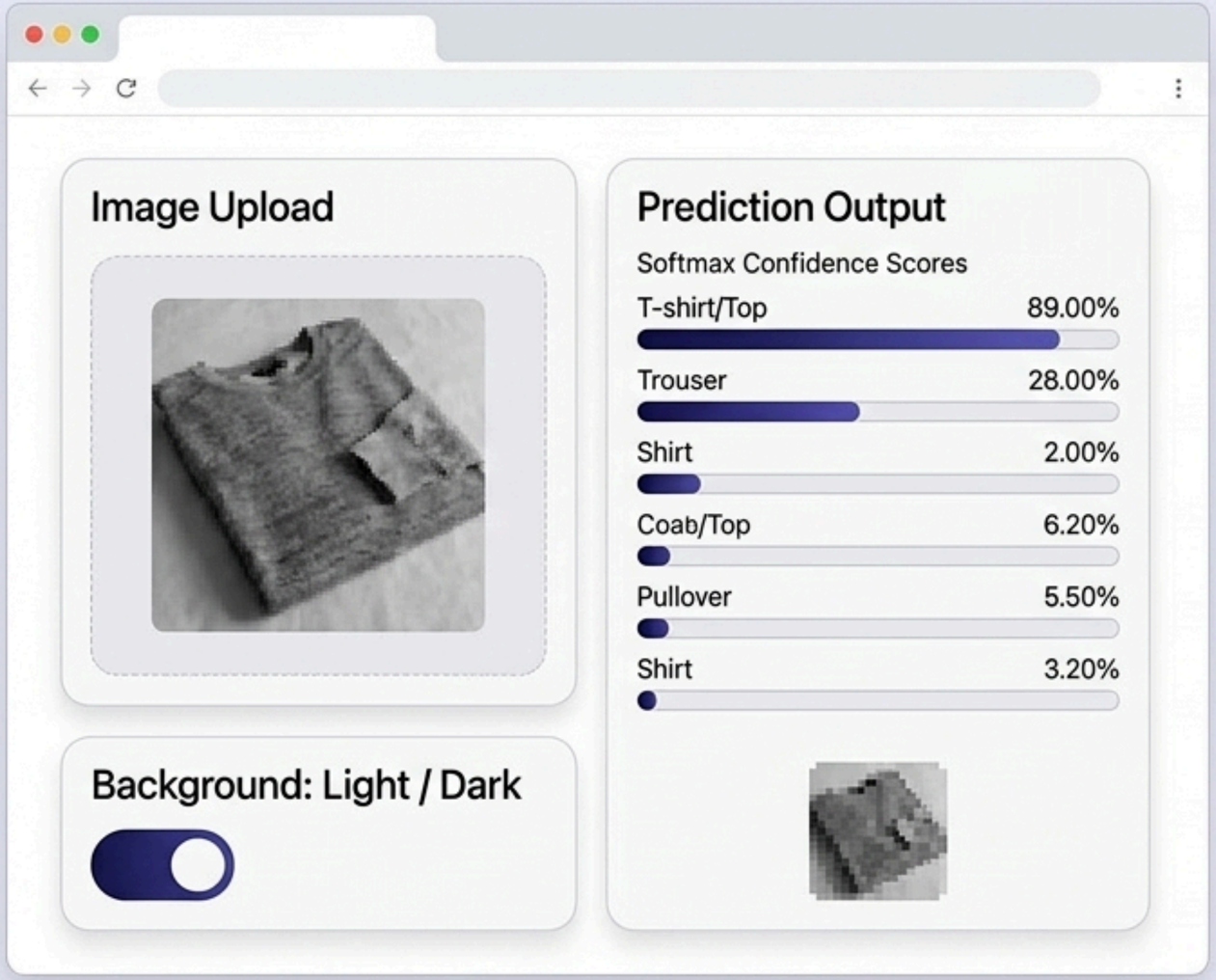
🏆 Best Classical Model: Random Forest (Raw Pixels). Accuracy: 85.27% | Macro F1: 85.09%.



Error Analysis: 'Shirts' proved the most difficult category to classify (52.9% recall). In low-resolution grayscale, the silhouette of a shirt heavily overlaps with pullovers, coats, and t-shirts.

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Deploying the Interface and Assessing Real-World Limitations



Interface & Architecture Note

Gradio app supports image upload, background toggling, and displays softmax confidence scores alongside the 28×28 processed input.

Crucial Clarity: The application demo is currently powered by a historical reference Convolutional Neural Network (CNN) achieving 89.00% accuracy, not the Phase 1 Classical Machine Learning models.

System Limitations

Real-world photos contain complex backgrounds, varying poses, and multiple objects, which degrades performance compared to the isolated, centered benchmark thumbnails.

Phase 1 Conclusion

Classical machine learning establishes a strong foundational baseline (85.27%). However, spatial architectures (like the reference CNN) are strictly necessary to push accuracy higher for production environments.

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